LISTING INFORMATION OF

Quad-Lock Insulating Concrete Forms (ICFs)

SPEC ID: 35720

Airfoam Industries Ltd. dba Quad-Lock Building Systems
19402 - 56 Ave
Surrey, BC V3S 6K4
Canada

LISTING INFORMATION

Quad-Lock ICFs are insulated concrete form (ICF) systems that are formed from expandable polystyrene (EPS) beads with polyethylene cross-ties and/or ABS fastening strips. The EPS panels are available in the following thicknesses:

- 2-1/4" Regular
- 3-1/8" Ultra
- 4-1/4" Plus

The ICF systems core nominal thicknesses are available in the following dimensions:

- 3-3/4"
- 5-3/4"
- 7-3/4"
- 9-3/4"
- 11-3/4"
- 13-3/4"

MATERIAL RATINGS

Test Standard	Test Type	Rating
ASTM C578	Physical Properties	Met Type II and IX Requirements
ASTM E2634	Physical Properties	Met Conditions of Acceptance
CAN/ULC-S701	Physical Properties	Met Type 2 and 3 Requirements
CAN/ULC-S717.1	Physical Properties	Met Conditions of Acceptance

CODE COMPLIANCE RESEARCH REPORT

Evaluation Method	Building Code	CCRR Number
ASTM E2634 ICC-ES AC353	2024, 2021, 2018, 2015, and 2012 International Building Code 2024, 2021, 2018, 2015, and 2012 International Residential Code	CCRR-1060

FIRE RATINGS

Test Standard	Rating	Design Number
ASTM E119 CAN/ULC-S101	2, 3, and 4 hours	QBS/ICF 240-01

Test Standard	Test Specimen	Rating
NFPA 286	EPS Panel Thickness: 4-1/4 in. 6 in. core Interior Finish: 1/2 inch regular gypsum board	Met Conditions of Acceptance

FLAME SPREAD RATINGS

UL723 SURFACE BURNING CHARACTERISICS ON EPS (CEILING VALUES REPORTED)

Material	Flame Spread Index	Smoke Developed Index
2.23 in. thick EPS	≤10	≤5

UL723 SURFACE BURNING CHARACTERISICS ON EPS (FLOOR VALUES REPORTED)

Material	Flame Spread Index	Smoke Developed Index
2.23 in. thick EPS	≤35	≤1150

This product is under the following quality assurance program:

Product	Product Approval No.
Quality-Lock Insulating Concrete Forms	FL 10431

Attribute	Value
Criteria	CAN / ULC S101 (2007)
Criteria	NFPA 286 (2011)
Criteria	CAN / ULC S717.1 (2012)
Criteria	ASTM E2634 (2011)
Criteria	ICC-ES AC353 (2012)
Criteria	ASTM C578 (2012a)
Criteria	UL 723 (2010)
Criteria	CAN / ULC S717.1 (2017)
Criteria	ASTM E119 (2020)
Criteria	ASTM E119 (2018b)
CSI Code	03 11 19 Insulating Concrete Forming
Listing Section	CONCRETE FORMS
Spec ID	35720

DRAWING INDEX

QBS/ICF 240-01

QBS/ICF 240-01

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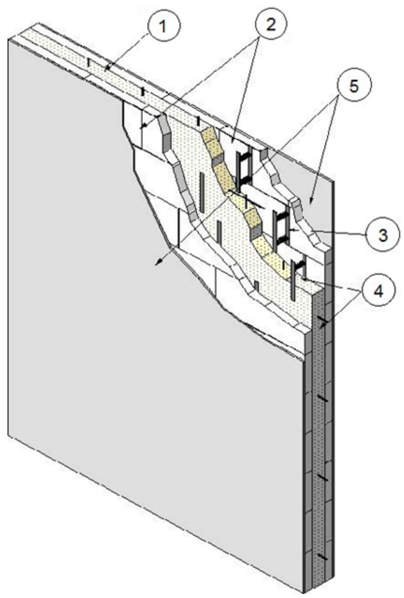
Division 03 – Concrete
03 11 00 Concrete Forming
03 11 19 Insulating Concrete Forming

Aqua-Pak Industries Ltd. dba Quad-Lock Building Systems
Design No. QBS/ICF 240-01
Insulating Concrete Forming
Quad-Lock Insulating Concrete Forms
ASTM E119
CAN/ULC-S101
Ratings: 2, 3, and 4 Hour



MIN. CORE THICKNESS	MAX. FIRE RATING
3-3/4 in. (96mm)	2 Hour
5-3/4 in. (146mm)	3 Hour
7-3/4 in. (197mm)	4 Hour

1. **CONCRETE:** Pour concrete with density of 150 lb/ft^3 (2400 kg/m^3) into the ICF system. The concrete shall have a min. 3000.0 psi (20.7 MPa) normal compressive strength after 28 days curing time, a max. slump of 6 in., max. aggregate size of 3/4 in. for core thicknesses greater or equal to 7-3/4 in., and 3/8 in. for core thicknesses less than 7-3/4 in. Place the steel reinforcement (not shown) before filling the ICF system with concrete. The rebar used is to be designed and placed per the applicable Code requirements and approved by a registered design professional with the appropriate license for the Authority Having Jurisdiction.

QBS/ICF 240-01 (2 OF 2)

Division 03 – Concrete
03 11 00 Concrete Forming
03 11 19 Insulating Concrete Forming

2. **CERTIFIED MANUFACTURER:** Aqua-Pak Industries Ltd. dba Quad-Lock Building Systems

   CERTIFIED PRODUCT: Quad-Lock Insulating Concrete Forms (ICFs)

   CERTIFIED MODEL: Regular: 2-1/4 in., Ultra: 3-1/8 in., and Plus: 4-1/4 in.

   The Quad-Lock ICF forming systems consist of Type II and IX molded expanded polystyrene (EPS) foam panels with embedded high density polyethylene cross-ties (Item 3) and/or ABS fastening strips (Item 4). The EPS panels used on Regular are manufactured from Type IX EPS beads; the Ultra and Plus panels use Type II EPS beads. The cross-ties are color coded to indicate different installation core thicknesses (cavity thicknesses). The Quad-Lock ICF core thicknesses are 3-3/4 in., 5-3/4 in., 7-3/4 in., 9-3/4 in., 11-3/4 in., and 13-3/4 in. The cross-ties are spaced by 12 in. intervals along the length of the constructed wall.

3. **HDPE CROSS-TIES:** Injection molded high density polyethylene cross-ties are 7-1/2 in., 9-1/2 in., 11-1/2 in., 13-1/2 in., 15-1/2 in., and 17-1/2 in. long; the cross-ties are color coded according to their lengths. Cross-ties are installed at the intersection of the horizontal and vertical joints of all panels and every 12 in. oc in between. The spacing is indicated by deep grooves in the EPS panels.

4. **ABS FASTENING STRIPS (Optional, Not Shown):** Fastening strips are available as an option for Regular and Plus panels. The fastening strips are 1-1/2 in. wide strips made from ABS plastic, and are molded into the EPS panels every 12 in. measured from the centre. The fastening strips provide anchoring for exterior wall cladding or siding.

5. **GYPSUM WALLBOARDS:** Non-fire-rated gypsum wallboards shall be installed on interior walls and the interior facing sides of exterior walls. Gypsum wallboards installed must have a min. thickness of 1/2 in. and min. area density of 1.54 lb/ft^2. The gypsum wallboards are to be fastened to the HDPE cross-ties, with #6 x 1-5/8 in. Type S drywall screws spaced 12 in. oc. Gaps between adjacent gypsum wallboards and screw head cavities shall be taped and filled with joint compound. The fire rating on exterior walls with gypsum wallboards installed on the interior facing side only applies from the inside.

6. **INTERIOR AND EXTERIOR FINISHES (Optional, Not Shown):** Interior and exterior finishes may be added to Quad-Lock ICF systems without affecting the fire-resistance rating.

   Exterior finishes may be applied to the exterior side of the forming system wall assembly without diminishing the assembly rating when desired. Exterior Insulation Finishing System (EIFS), any exterior stucco, brick or brick veneer, stone or stone veneer, cultured stone and siding made from vinyl, aluminum, wood, or steel may be used. Apply exterior finishes in accordance with the manufacturer's instructions.

7. **WALL ASSEMBLY:** The Quad-Lock ICF system may be used as either an interior or exterior wall. ICFs exposed to the interior of a building shall have a thermal barrier attached. Exterior walls are only required to have a thermal barrier on the side facing the interior of the building. The fire resistance is applicable to the Quad-Lock ICF systems from either side.

Code Compliance Research Report
CCRR-1060

Issue Date: 05-06-2016
Revision Date: 04-29-2026
Renewal Date: 04-30-2027

DIVISION: 03 00 00 – CONCRETE
Section: 03 11 19 – Insulating Concrete Forming

REPORT HOLDER:
Airfoam Industries Ltd. dba Quad-Lock Building Systems
19402 – 56 Avenue
Surrey, B. C. V3S 6K4
800-663-8162
www.airfoam.com

REPORT SUBJECT:
Quad-Lock Insulating Concrete Forms (ICFs)

1.0 SCOPE OF EVALUATION

1.1 This Research Report addresses compliance with the following Codes:

- 2024, 2021, 2018, 2015, 2012 *International Building Code®* (IBC)
- 2024, 2021, 2018, 2015, 2012 *International Residential Code®* (IRC)

NOTE: This report references 2024 Code sections. Section numbers in earlier versions of the Code may differ.

1.2 Quad-Lock ICFs have been evaluated for the following properties:

- Physical properties
- Surface-burning characteristics
- Fire resistance
- Structural
- Use in Type V construction
- Exterior walls in Types I through IV construction
- Attic and crawl space fire evaluation

See Table 1 for applicable Code sections related to these properties.

1.3 Quad-Lock ICFs have been evaluated for the following uses:

- Use as stay-in-place formwork for structural concrete, loadbearing and non-loadbearing exterior and interior walls, concrete beams, lintels, foundation walls, and retaining walls in Type V construction
- Use in fire-resistance-rated construction, provided installation is in accordance with Section 4.9
- Use in Types I, II, III, or IV construction when installed in accordance with Section 4.10

2.0 STATEMENT OF COMPLIANCE

Quad-Lock Insulating Concrete Forms (ICFs) comply with the Codes listed in Section 1.1, for the properties stated in Section 1.2 and uses stated in Section 1.3, when installed as described in this report, including the Conditions of Use stated in Section 5.

2.1 2024 IBC and IRC Evaluation Reports: The Intertek CCRR is an *Evaluation Report* for approval of an alternate material, design, or method of construction in accordance with Section 104.2.3.6.1 of the 2024 IBC and Section R104.2.2.6.1 of the 2024 IRC.

3.0 DESCRIPTION

3.1 General: The Quad-Lock ICFs consist of two expanded polystyrene (EPS) foam plastic panels separated by injection-molded high-density polyethylene (HDPE) cross-ties which are inserted into top and bottom edges of the EPS panels. Quad-Lock ICFs are shipped as components and are assembled on the jobsite.

The HDPE cross-ties maintain the EPS panels at fixed clear distances outlined in Section 3.3. Galvanized metal corner brackets are provided to reinforce corners, T-junctions and angles. Galvanized metal tracks are provided for optional use at the base and top of the walls to provide form alignment and resist concrete pressure. Metal wire top ties for optional use connect the panels with metal tracks at the top of the walls. See Figures 1 and 3 for illustration of the forms.

Quad-Lock forms are classified as a flat ICF wall system in accordance with IRC Section R608.3.1.

3.2 Foam Plastic Panels: The EPS panels are 12 in. high, 48 in. long. Quad-Lock Regular Panels are 2-1/4 in. thick. Ultra Panels are 3-1/8 in. thick and Plus Panels are 4-1/4 in. thick. The Regular Panels have a nominal density of 1.9 pcf. Ultra and Plus Panels have a nominal density of 1.44 pcf. All panels have a flame-spread index of 25 or less and a smoke-developed index of 450 or less when tested in accordance with ASTM E84, and comply with ASTM C578 as Type II (for Ultra and Plus Panels) and Type IX (for Regular Panels). The EPS panels have pre-formed slots to receive tie flanges at both top and bottom edges and preformed interlock knobs to facilitate positioning of panels while stacking.

3.3 Cross-Ties: Quad-Lock HDPE cross-ties each consist of 4 tie flanges and two legs that span across the concrete cavity. The legs are joined to one another, 4 in. apart, to form a paired assembly. During assembly on-site, cross-ties are spaced at 12 in. on center along horizontal joints between panels, embedding half of each tie-flange into panels below. The panel above receives the upper half of tie flanges into molded slots at the bottom of the panel. The face of each tie flange is recessed from the outer face of EPS panels by 3/8 in. Legs of cross-ties have regular depressions along their top edge to assist in positioning horizontal reinforcing bars. Cross-ties are color-coded by length to assist in choosing the appropriate concrete cavity size. Tie color tints are black, blue, yellow, green, red, and brown. When used with Regular and Plus Panels cross-ties form cavities of 3-3/4 in., 5-3/4 in., 7-3/4 in., 9-3/4 in., 11-3/4 in., or 13-3/4 in. When used with Ultra Panels, cross-ties maintain clear distances of 4 in., 6 in., 8 in., 10 in., and 12 in. Quad-Lock XT-Extender Ties are available to increase the standard wall widths by 12 in. and are tinted orange.

Optional ABS fastening strips are available to provide anchoring for exterior wall cladding or sidings. The ABS fastening strips are 1-1/2 in. wide by 10-7/8 in. tall and are molded into the EPS panels every 12 in. on center.

Each flange is 1-1/2 in. wide by 4-7/8 in. high by 1/8 in. thick. Cross-tie flanges interlock with tops and bottoms of ABS fastening strips as each row is added.

3.4 Concrete: Concrete must be normal-weight concrete complying with the applicable Code and must have a maximum aggregate size of 3/4 in. for core thicknesses greater or equal to 7-3/4 in., and 3/8 in. for core thicknesses less than 7-3/4 in., and a minimum specified compressive strength of 2500 psi at 28 days for non-fire-resistance-rated construction, and 2900 psi for fire-resistance-rated construction. Under the IRC, concrete must comply with IRC Section R404.1 (foundation walls and retaining walls) and R608.5.1 (walls), as applicable.

3.5 Reinforcement: Deformed steel reinforcement bars must have a minimum specified yield stress of either 40 ksi or 60 ksi, depending on the structural design. Under the IBC, the deformed steel bars must comply with Section 3.5.3.1 of ACI 318 and IBC Section 1903. If construction is based on the IRC, reinforcement must comply with IRC Sections R404.1.3.3 and R608.5.2.

4.0 DESIGN AND INSTALLATION

4.1 General: Design and installation of Quad-Lock ICFs must comply with this report, the applicable Code, and the manufacturer's published installation instructions, which must be available at the jobsite at all times during installation.

4.2 Design:

4.2.1 IBC Method: Solid concrete walls must be designed and constructed in accordance with IBC Chapter 16 and 19, as applicable. Footings and foundations must be designed in accordance with IBC Chapter 18.

4.2.2 Alternative IBC Wind Design Method: Solid concrete walls may be designed and constructed in accordance with the provisions of Section 209 of ICC 600, subject to the limitations found in Exception 1 of IBC Sections 1609.1.1 and 1609.1.1.1. Design and construction under the provisions of ICC 600 are limited to resisting wind forces.

4.2.3 IRC Method: Solid concrete walls, footings, and foundations must be designed in accordance with IRC Sections R608 and R404.1.3, as applicable for flat wall systems.

4.2.4 Alternative IRC Methods: When used to construct buildings that do not conform to the applicability limits of IRC Sections R404.1.3 and R608, construction must be in accordance with the prescriptive provisions of the 2017 Prescriptive Design of Exterior Concrete Walls (PCA 100), or 2012 PCA 100 for 2015 and 2012 IRC, or the structural analysis and design of the concrete must be in accordance with ACI 318 and IBC Chapters 16, 18, and 19.

4.3 Wall Construction: The Quad-Lock ICF wall system must be supported on concrete footings complying with IBC Chapters 18 and 19, or IRC Chapter 4, as applicable.

Vertical reinforcement bars embedded in the footing must extend into the base of the wall system the minimum development length necessary for compliance with ACI 318, Chapter 12 (IBC) or IRC Section R608.5.1, as applicable. Vertical and horizontal reinforcement bars must have concrete protection in accordance with, and must be placed as required by, the design and the applicable Code. Additional reinforcement around doors and windows must be described in the approved plans. The panels must be installed according to the manufacturer's instructions, with the cross-ties placed in each row and spaced at a maximum of 12 in. on center. The ties must be vertically aligned in each row to support the interior and exterior finish materials. Placement of the panels must begin from any two corners proceeding to fill in between the corners to form an individual wall section.

Concrete quality, mixing, and placement must comply with IBC Chapter 19 or IRC Sections R404.1.3.3 and R608.5.1, as applicable. Window and door openings must be built into the forms, with the same dimensions as the "rough stud opening" specified by the window or door manufacturer, prior to the placement of the concrete. Connections of concrete walls to footings, floors, ceilings and roofs must be in accordance with IRC Section R608.9, or be engineered in accordance with the IBC, whichever Code is applicable. Anchor bolts used to connect wood ledgers and plates to the concrete must be cast in place, with the bolts sized and spaced as required by design and the applicable Code. Details must be prepared to accommodate the specific job situation, in accordance with the applicable Code and the requirements of this report, subject to the approval of the Code official.

4.4 Interior Finish:

4.4.1 General: ICF units exposed to the building interior must be finished with an approved 15-minute thermal barrier, such as minimum 1/2 in. thick regular gypsum wallboard complying with ASTM C1396, installed vertically or horizontally and attached to the cross-tie flanges with minimum 1-5/8 in. long, No. 6, Type W, coarse-thread gypsum wallboard screws spaced 12 in. on center vertically and 12 in. on center horizontally. The screws must penetrate a minimum of 1/4 in. through the flange. Gypsum board joints and screw heads must be taped and finished with joint compound in accordance with ASTM C840 or GA216.

4.4.2 Attic and Crawl Space Installations: When the ICFs are used for walls of attic or crawl spaces, an ignition barrier complying with IBC Section 2603.4.1.6 or IRC Sections R303.5.3 or R303.5.4, is required, except when all of the following conditions are met:

- Entry to the attic and crawl space is only to service utilities, and no storage is permitted
- There are no interconnected attic or basement areas
- Air in the attic or crawl space is not circulated to other parts of the building
- Under-floor (crawl space) ventilation is provided that complies with IBC Section 1202.4 or IRC Section R408.1, as applicable
- Attic ventilation is provided when required by IBC Section 1202 or IRC Section R806, as applicable
- Combustion air is provided in accordance with IMC (International Mechanical Code) Section 701
- The ICFs must have at least one label as described in Section 7.0 visible in every 160 square feet of wall area

4.5 Exterior Finish:

4.5.1 Above Grade: The exterior surface of the ICF must be covered with an approved wall covering in accordance with the applicable Code or a current Research Report. Under IRC, walls must be flashed in accordance with IRC Section R703.4. When the wall covering is mechanically attached to structural members, the wall covering must be attached to the flanges of the cross-ties or FS Strips with fasteners described in Table 3, having sufficient length to penetrate through the flange a minimum of 1/4 in. The fasteners have an allowable fastener withdrawal and lateral shear strength as noted in Table 3.

The fastener spacing must be designed to support the gravity loads of the wall covering and to resist the negative wind pressures.

The negative wind pressure capacity of the exterior finish material must be the same as that recognized in the applicable Code for generic materials, or that recognized in a current evaluation report for proprietary materials and must not exceed the maximum withdrawal capacity of the fasteners listed in Table 3.

4.5.2 Below Grade: For below-grade applications, exterior wall surfaces must be dampproofed or waterproofed in accordance with IBC Section 1805 or IRC Section R406, as applicable. The material must be compatible with the ICF foam plastic units, and free of solvents, hydrocarbons, ketones, and esters that will adversely affect the EPS foam plastic panels. No backfill can be applied against the wall until the complete floor system is in place, unless the wall is designed as a freestanding wall that does not rely on the floor system for structural support.

4.6 Foundation Walls: The ICF system may be used as a foundation stem wall provided the structure is supported on concrete footings complying with the applicable Code. For jurisdictions adopting the IRC, compliance with Section R404 is required.

4.7 Retaining Walls: The ICF system may be used to construct retaining walls, with reinforcement designed in accordance with accepted engineering principles, Section 4.2 of this report, and the applicable Code.

4.8 Protection Against Termites: Where the probability of termite infestation is defined by the Code official as "very heavy", the foam plastic must be installed in accordance with IBC Section 2603.8 or IRC Section R305.4, as applicable. Areas of very heavy termite infestation must be determined in accordance with IBC Figure 2603.8 or IRC Figure R305.4.

4.9 Fire-resistance-rated Construction: Walls constructed with Quad-Lock ICFs have fire-resistance ratings for bearing and nonbearing wall assemblies as shown in Table 2.

4.10 Installation in Buildings Required to be of Types I, II, III, and IV Construction:

4.10.1 General: Exterior walls constructed with the ICFs for use in buildings required to be of Type I, II, III, or IV construction must comply with the applicable conditions cited in Sections 4.10.2 through 4.10.4.

4.10.2 Interior Finish:

4.10.2.1 Buildings of Any Height: The ICFs must be finished on the interior with an approved 15-minute thermal barrier, such as 1/2 in. thick gypsum wallboard, as required by the IBC. The gypsum wallboard must be installed and attached as described in Section 4.3.1.

4.10.2.2 Alternate Interior Finish for One-story Buildings: For one-story buildings, the interior finish may be in accordance with IBC Section 2603.4.1.4, provided all the conditions in that section are met.

4.10.3 Exterior Finish:

4.10.3.1 Buildings of Any Height: Except as allowed in Section 4.5.1, the ICFs must be finished on the exterior with materials described in Sections 4.10.3.1.1, 4.10.3.1.2, or 4.10.3.1.3. The ICFs must have at least one label as described in Section 7.0 visible in every 160 square feet of wall area prior to applying the wall covering.

4.10.3.1.1 Exterior Finish – EIFS and One-coat Stucco: EIFS and one-coat stucco wall coverings may be applied over the Quad-Lock ICFs, provided the wall covering system is recognized in a current Research Report and is recognized for use in Types I, II, III, and IV construction. The wall covering system must be installed in accordance with the respective Research Report and the maximum mass of foam plastic per wall surface area [lbs/ft^2] qualified in the wall covering evaluation report must be greater than 0.312 lbs/ft^2 (which is the mass of the EPS panel on the exterior side of the concrete wall). Acceptable EIFS wall coverings include the following:

- STO Corp., StoTherm Classic NexT EIFS, ICC-ES ESR-1748
- STO Corp., StoTherm Classic, StoTherm Essence, and StoTherm Premier Systems, ICC-ES ESR-1720
- Master Builders Solutions US, LLC, Senerflex Wall System EIFS, ICC-ES ESR-1878
- Dryvit Systems, Inc., Dryvit Outsulation System, ICC-ES ESR-1232

Ignition properties in accordance with NFPA 268 must be provided to the Code official for the specific EIF system, as required by IBC Section 2603.5.7.

4.10.3.1.2 Exterior Finish – Brick Veneer: Anchored brick veneer must be attached to the exterior face of Quad-Lock ICF walls with approved masonry anchors extending into the concrete as required in the IBC. The 4 in. thick brick veneer must comply with the IBC and must be installed with a minimum 1 in. air gap between the face of the exterior EPS panel and the brick. The brick must be installed with a steel shelf angle attached to the concrete and installed at each floor line and at the top of each window and door opening. The assembly satisfies the requirements for ignition under the exceptions to IBC Section 2603.5.7.

4.10.3.1.3 Exterior Finish – Plaster: Metal lath and exterior plaster or standard stucco shall comply with the requirements of the IBC and the exterior plaster must have a minimum thickness of 7/8 in. The lath shall be attached to the cross-tie flanges with fasteners as described in Section 4.4.1. The assembly satisfies the requirements for ignition under the exceptions to IBC Section 2603.5.7.

4.10.3.1.4 Other Exterior Wall Coverings: Other wall coverings must be demonstrated to the satisfaction of the building official as meeting the requirements of IBC Section 2603.5. Assemblies tested in accordance with NFPA 285 must include EPS having a maximum mass per wall surface area (in lbs/ft2) no greater than 0.312 lbs/ft2 (1.52 kg/m2) (which is the maximum-tolerance mass of the EPS panel on the exterior side of the concrete wall).

4.10.4 Fireblocking: For applications on buildings of any height, floor-to-wall intersections must be fireblocked in accordance with the IBC to prevent the passage of flame, smoke and hot gases from one story to another. The foam plastic on the interior side of the exterior walls and on both sides of interior walls must be discontinuous from one story to another. See Figure 4, Floor Connections for Type I - IV Construction (Typical). Details of typical floor-to-wall intersections must be shown on approved drawings.

4.11 Special Inspection:

4.11.1 IBC: Special inspection is required as noted in IBC Section 1705 for placement of reinforcing steel and concrete, and for concrete cylinder testing.

When an EIFS wall covering is applied, special inspection is required in accordance with the evaluation report on the EIFS and with IBC Sections 1704 and 1705.16 is required.

4.11.2 IRC: For walls designed in accordance with Section 4.2.3 or PCA 100, special inspection is not required. When walls are designed in accordance with the IBC, as described in Section 4.2.4, special inspection is required as described in Section 4.11.1.

5.0 CONDITIONS OF USE

The Quad-Lock Building Systems Ltd. Insulating Concrete Forms described in this Research Report comply with, or are suitable alternatives to, what is specified in those Codes listed in Section 1.0 of this report, subject to the following conditions:

5.1 The ICFs must be manufactured, identified and installed in accordance with this Research Report, the manufacturer's published installation instructions, and the applicable Code. In the event of a conflict between the manufacturer's instructions and this report, this report governs.

5.2 When required by the Code official, calculations showing compliance with the general design requirements of the applicable Code must be submitted to the building official for approval, except where calculations are not required under IRC Section R608.1. The calculations and details must be prepared by a registered design professional where required by the statutes of the jurisdiction in which the project is to be constructed.

5.3 When required by the Code official, calculations and details showing compliance with IRC Section R608.5.3 and R404.1.3.3.6 must be submitted, establishing that the ICFs provide sufficient strength to contain concrete during placement and the cross-ties are capable of resisting the forces created by fluid pressure of fresh concrete. The calculations and details must be prepared by a registered design professional where required by the statutes of the jurisdiction in which the project is to be constructed.

5.4 The ICFs must be separated from the building interior with an approved 15-minute thermal barrier.

5.5 Use of the ICF system in Types I, II, III, and IV construction must be as described in Section 4.10.

5.6 The plastic cross-ties must be stored indoors away from direct sunlight.

5.7 Special inspection must be provided in accordance with Section 4.11 of this report.

5.8 The Quad-Lock ICFs are manufactured in Surrey, British Columbia, Canada, and Villa Rica, Georgia, USA, under a quality control program with inspections by Intertek Testing Services NA, Inc.

6.0 SUPPORTING EVIDENCE

6.1 Reports of tests in accordance with ASTM E119 and ASTM E2634.

6.2 Data in accordance with the ICC-ES Acceptance Criteria for Stay-In-Place, Foam Plastic Insulating Concrete Forms (ICF) Systems for Solid Concrete Walls (AC353), dated October 2012, editorially revised 2021.

6.3 Intertek Listing Report "Quad-Lock Insulating Concrete Forms (ICFs)".

7.0 IDENTIFICATION

The Quad-Lock Insulating Concrete Forms are identified by a label bearing the report holder's name (Airfoam Industries Ltd. dba Quad-Lock Building Systems), the manufacturing location, the lot number, the Intertek Mark, and the Code Compliance Research Report number (CCRR-1060).

When use is in buildings required to be of Type I, II, III, or IV construction, one label must be visible in every 160 square feet of wall area.

When the forms are used in an attic or crawl space without an ignition barrier, the exposed inside face of the ICF must be labeled with the phrase "Acceptable for use in attics and crawl spaces". The label must be visible in every 160 square feet of wall area.

8.0 OTHER CODES

This section is not applicable.

9.0 CODE COMPLIANCE RESEARCH REPORT USE

9.1 Approval of building products and/or materials can only be granted by a building official having legal authority in the specific jurisdiction where approval is sought.

9.2 Code Compliance Research Reports shall not be used in any manner that implies an endorsement of the product by Intertek.

9.3 Reference to the https://bpdirectory.intertek.com is recommended to ascertain the current version and status of this report.

TABLE 1 – PROPERTIES EVALUATED

PROPERTY	APPLICABLE CODE SECTIONS[1]	
	IBC	IRC
Physical properties	NA	R404.1.3.3.6.1
Surface Burning Characteristics	2603.3	R303.3
Fire resistance	703.2	R302.1
Structural	Chapter 19	Section R610
Exterior walls in Types I – IV construction	2603.5	NA
Attic and crawl space applications	2603.4.1.6 and 2603.9	R316.5.3, R316.5.4 and R316.6

[1]Section numbers pertain to the most recent edition cited in Section 1.1 of this Report

TABLE 2 – TWO, THREE, OR FOUR-HOUR FIRE-RESISTANCE-RATED WALL ASSEMBLIES[4]

WALL TYPE	FIRE RESISTANCE	MINIMUM CONCRETE WIDTH	THERMAL BARRIER	STEEL REINFORCEMENT
Load Bearing[1, 2,3,4,5] Max load 29,800 lbf/lin ft	2-Hour	3-3/4 in.	½ in. gypsum wallboard fastened 12 in. oc in field and 12 in. oc at perimeter	Vertical - #4 at 15-3/4 in. oc Horizontal - #4 at 12 in. oc
Load Bearing[1,2,,3,4,5] Max load 29,800 lbf/lin ft	3-Hour	5-3/4 in.	½ in. gypsum wallboard fastened 12 in. oc in field and 12 in. oc at perimeter	Vertical - #4 at 15-3/4 in. oc Horizontal - #4 at 12 in. oc
Load Bearing[1,2,3,4,5,] Max load 29,800 lbf/lin ft	4-Hour	7-3/4 in.	½ in. gypsum wallboard fastened 12 in. oc in field and 12 in. oc at perimeter	Vertical - #4 at 15-3/4 in. oc Horizontal - #4 at 12 in. oc
Nonbearing[2,3,4]	2-Hour	3-3/4 in.	½ in. gypsum wallboard fastened 12 in. oc in field and 12 in. oc at perimeter	Vertical - #4 at 15-3/4 in. oc Horizontal - #4 at 12 in. oc
Nonbearing[2,3,4]	3-Hour	5-3/4 in.	½ in. gypsum wallboard fastened 12 in. oc in field and 12 in. oc at perimeter	Vertical - #4 at 15-3/4 in. oc Horizontal - #4 at 12 in. oc
Nonbearing[2,3,4]	4-Hour	7-3/4 in.	½ in. gypsum wallboard fastened 12 in. oc in field and 12 in. oc at perimeter	Vertical - #4 at 15-3/4 in. oc Horizontal - #4 at 12 in. oc

[1]Steel reinforcement is the minimum required for the design loads given.
[2]Concrete must be normal-weight concrete (150-155 pcf) with a minimum 2900 psi compressive strength.
[3]Fasteners to attach the gypsum wallboard thermal barrier must be 1-5/8-inch long No. 6, Type W, coarse-thread gypsum wallboard screws. See Section 4.4.1
[4]The wall assembly may be used as either an interior or exterior wall. When used as an interior wall, both sides of the form must be protected with gypsum wallboard.
[5]Design loads are based on 10-foot wall heights.

TABLE 3 – ALLOWABLE WITHDRAWAL AND LATERAL CAPACITIES OF FASTENERS IN CROSS-TIE FLANGES

Lateral (Shear) Strength Evaluation Summary		
Sample Designation		**Allowable Lateral Strength (lb_f)**
Substrate	**Fastener Type**[1]	
Cross-Tie Flanges	#9 Cement Board Fasteners	46
	#8 Drywall Screws	39
FS Fastening Strips	#9 Cement Board Fasteners	56
Withdrawal Strength Evaluation Summary		
Sample Designation		**Allowable Withdrawal Strength (lb_f)**
Substrate	**Fastener Type**[1]	
Cross-Tie Flanges	#9 Cement Board Fasteners	24
	#8 Drywall Screws	24
FS Fastening Strips	#9 Cement Board Fasteners	36

[1]Fasteners must be of sufficient length to penetrate through the flange a minimum of 1/4 inch

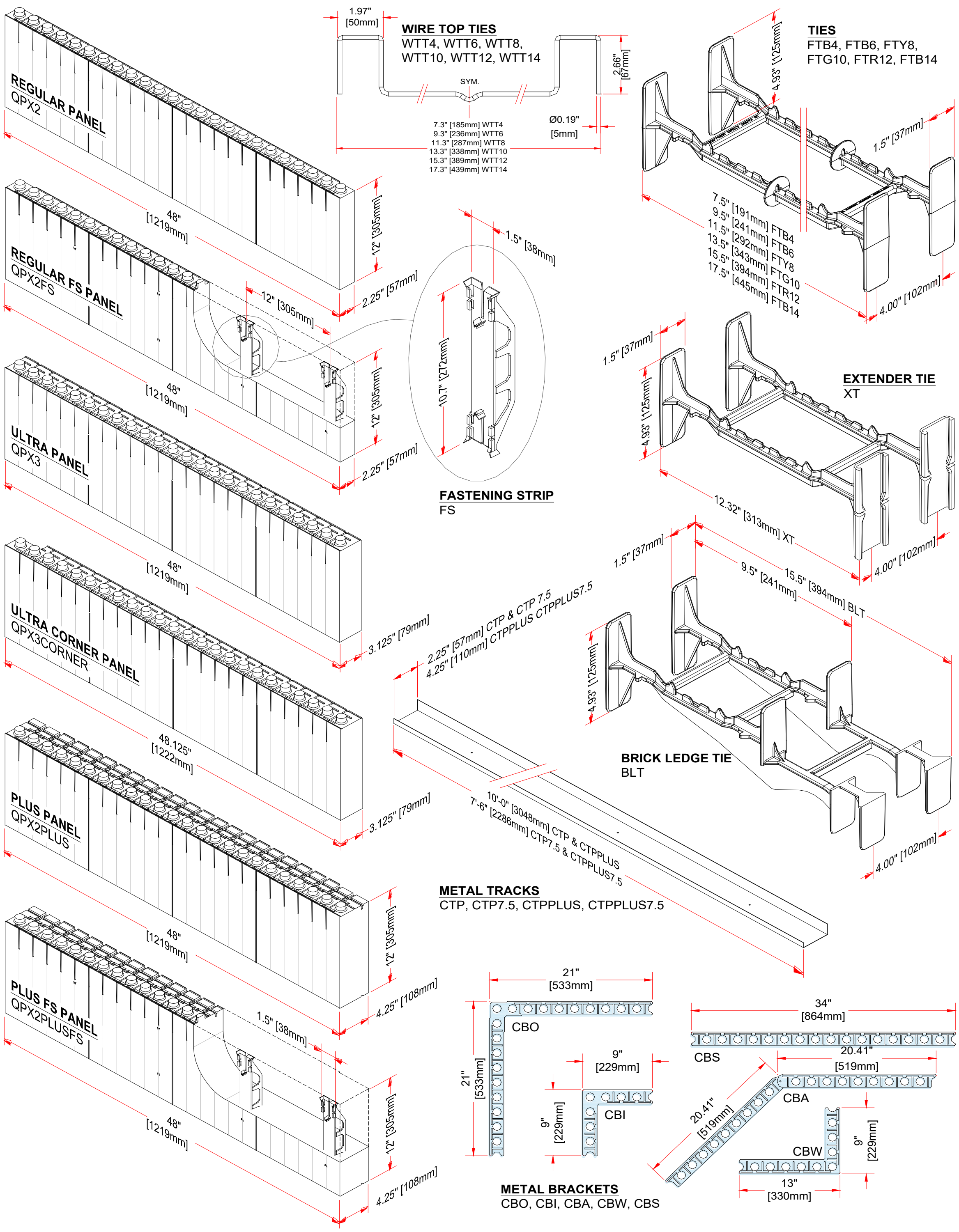
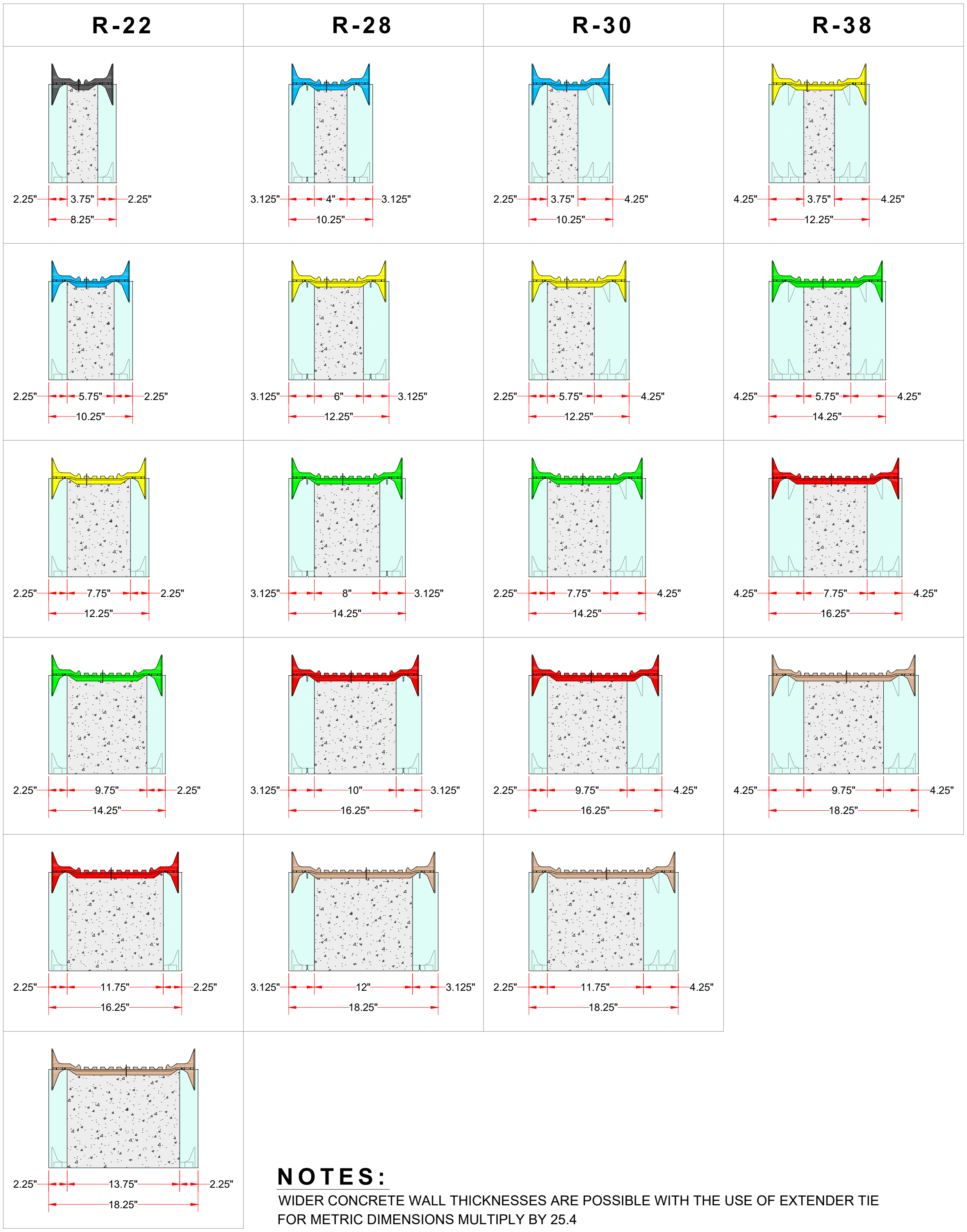


FIGURE 1 – QUAD-LOCK FORMS AND ACCESSORIES



NOTES:

WIDER CONCRETE WALL THICKNESSES ARE POSSIBLE WITH THE USE OF EXTENDER TIE

FOR METRIC DIMENSIONS MULTIPLY BY 25.4

FIGURE 2 – QUAD-LOCK ICF WALL CONFIGURATIONS

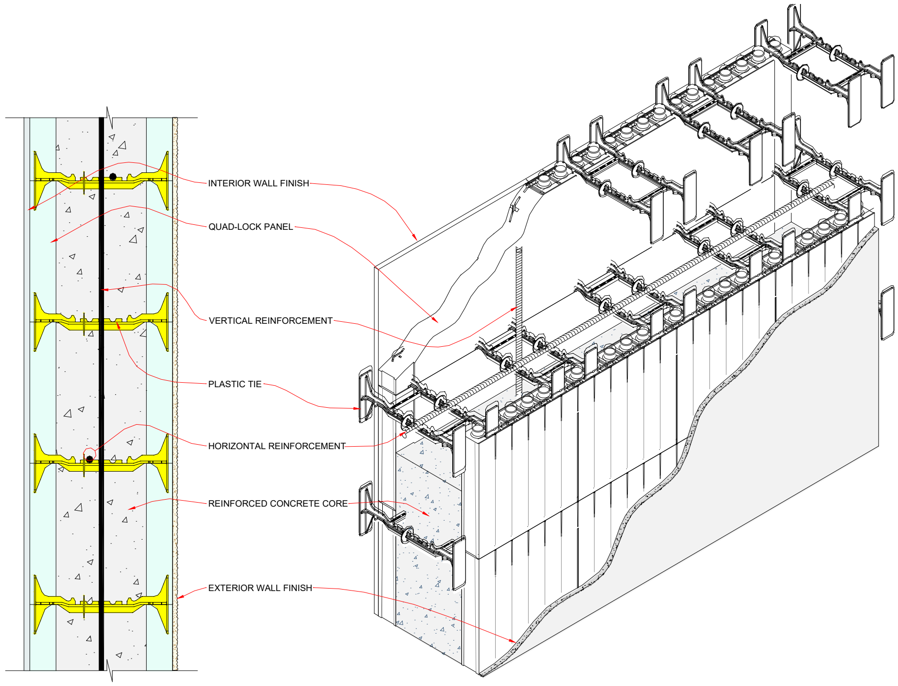


FIGURE 3 – QUAD-LOCK ICF WALL SYSTEM

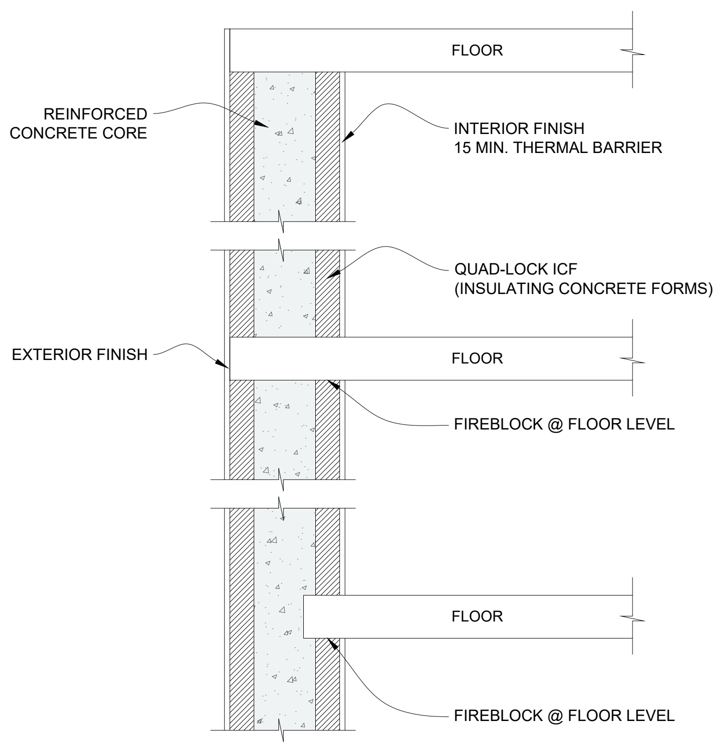


FIGURE 4 – FLOOR CONNECTIONS FOR TYPES I – IV CONSTRUCTION